IR CHECKLIST

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1. BASIC INFORMATION	
Process	
Project name	
Customer	
Application	
Project manager	
IP design classificaiton	Ne
QA manager	
Target Design review	
MTO Schedule	

2. DESIGN CHECK SHEET

CATEGORY	ITEM
Customer Request	Process - Metal Stack
Customer Request	Process - PDK Version
Customer Request	Function - SGMII Support
Customer Request	Function - Transmission Mode
Customer Request	Function simulation
Customer Request	Function simulation
Customer Request	Function simulation

Customer Request	**Function simulation**
Customer Request	**Function simulation**
Customer Request	**Function simulation**
Customer Request	**Function simulation**
Customer Request	**Function simulation**
Fundamental information	**Temperature & Reliability check condition**
IP Version check	**IP DK Version**
	HTOL
	Power up check
	Start up check
	Power down check
	Function simulation
	Function simulation
	Function simulation

Function check	**Function simulation**
	SFR Setting Check
	Function - ATE(Automatic Test Equipment) Vector Test
	Function simulation
	Function simulation
	Unused pin check
	APB _ Write Transfer
	APB _ Read Transfer
Implementatiaon	**Timing Skew Check for Direct Access Port in Test Mode**
	STA(Static Timing Analysis)

check	**Reference Clock**
	Network Jitter
	Product floor plan
	PAD
	PAD
	PAD

Floor plan	**Power & Ground Routing**
	ESD Protection (I/O Information)
	ESD Protection (CDM Diode insertion)
	ESD Protection (Back to Back diode connection)
	IP Placement
	IP Placement
Verification check	**Device List check**
	Package Ground Plane Merge

Off-Chip	**Supply constraints**
	Package Sign-Off
	Ballmap

SF 5nm LPE	
Dolphin 7	
Telechips	
w design / Modification	

CHECK POINT	
Is the metal stack (13M_4Mx_7Dx_2Iz_LB) finalized and GDS-level verified?	Confirm layout GDS ve
Is the PDK version confirmed and used consistently across schematic, layout, and signoff	Cross-check with DK re
Does the IP support all required SGMII speeds (10/100/1000/2500 Mbps)?	Confirm specification c
Does the IP support both full and half duplex operation?	Review functional simu
Are the internal SerDes blocks properly implemented and functionally verified?	Confirm RTL implemen
Is internal PLL properly implemented and lock behavior validated?	Confirm with PLL spec
Is PCS logic included and operation per spec?	Review functional bloc control.

Are the GMII and APB interfaces functionally verified and compliant with IP-SoC spec?	Confirm GMII data trar
Is SGMII external interface verified for interoperability and timing requirements?	Confirm through integ
Is the reference clock (24 MHz / 156.25 MHz) selection logic implemented correctly and verified?	Review RTL logic, cloc
Is ETH_CLK (up to 312.5MHz) properly generated and synchronized with system GMAC?	Confirm output freque
Is 156.25MHz differential output correctly generated from SERDES internal PLL and compliant with spec for external PHY usage?	Confirm clock output s
Is the chip operating temperature condition and life cycle checked?	Review and make com - Chip operating envir - Junction temperatur - The reliability(EM) cy
Check whether the version of the DB used is final (FE/BE/DOC)	Have you used the late Enter the name of the
In full-chip level HTOL(SCAN) test mode, have you checked that the IP's inputs are set as conditions guaranteed by the UserGuide? Have you verified that the IP operates without any problem by simulation	1) Please share dump 2) Please share the IP' 3) Please share used w
Is power-management sequence which is supplied to IP guaranteed in SoC?	Please share the figure - If you share a dump f - If there is no figure o etc.).
IP start-up sequence check	Have you checked tha described in the Datas refer to the "1.Power s
Power down sequence check	Have you checked tha the Datasheet?
Input control sequence during the operation	Have you checked If th operation? Is there any real integration enviro
Are the verification results in product level shared with IP designer? (IP normal scenario; RTL simulation or post simulation results under the actual product scenario) Please check whether appropriate control and default value are applied through those Verilog simulations.	1) Please share dump 2) Please share the IP' 3) Please share used w
Stand-by scenario during lock-time	If the IP is a clock IP w is reflected to the syst in the Datasheet?

Test mode verification	Have you checked tha output to I/O is within Write the maximum fr
Are SFR setting values applied as the guide which is provided by IP designer?	Please reviews it and a
Are function test results reviewed? (All about IP's test mode) Please check whether appropriate control and default value are applied through those Verilog simulations.	1) Please share dump 2) Please share the IP' 3) Please share used v
Input controllability	Have you checked if th cannot be easily modif If there is, write the lis
Input controllability	Have you verified the no problem?
Are there any IP input/output pins that are not used? All IP inputs except the unused-pin list must be controllable.	3) If all pins are contro charge. • On chip guide -. The unused input pi operation) or the po -. The unused output p • Off chip guide
Check for write sequence	Has the write transfer • PADDR, PWRITE, P • After the following • PADDR, PWDATA, at the end of the cycle • PENABLE signal is • Only one signal sh
Check for read sequence	Has the read transfer • PADDR, PWRITE, a • After the following • The slave provide • The data is sampl
Are timing skew for ports which access directly from GPIO checked for direct access (stand-alone) test? When in PHY Test mode, the APB interface is accessed immediately from the GPIO. At this time, check whether skew is considered for Multi bit. (For example, PADDR (16bit), RDATA, PWDATA (16bit), PSEL, PENALBE, PWRITE) If there is a difference in skew, eventually the ATE must turn on the timing to skew. In fact, it is difficult to consider skew in the ATE. In the Post STA result, the skew report is received separately.	Please attach the rep between ports for ATE Use skew report from Please, refer IP user gu
Are STA results between IP and product reviewed? 1) Review the unconstraint path with the IP designer (The important timing path must be constraint path) 2) Check the timing report (setup / hold timing reports)	1) Please share STA Re STA (e.g. FIN, FOUT) 2) Plase share the unc

Check for reference clock jitter	Jitter in the reference c jitter characteristics. Ji source. If you use an internal c clock- tree network sh condition). Otherwise, NOTE: It is mandatory estimated reference cl
Are the clock network jitters from full chip shared and reviewed with IP designers? -. check for maximum jitter	Please attach the resu pin -. Check the input clco jitter before product
Is the floorplan shared with IP designer in advance?	rs_etphy5000r_ln05lpe cannot be placed in th The PHY IP is also recc blocks is recommende If there are noisy block side). The noisy switch package/board level. T in the die. For the process monitc recommend you to pla of the PHY (e.g.< 400 from our PHY (e.g. 50
I/O pad type	Have you checked tha
Isolation of I/O pad	Is the Power I/O for the reviewed over any ESD
Noise around I/O pad	Have you checked tha

check for prevention of supply noise	In order to prevent pov rs_etphy5000r_ln05lpe
Refer to the check method	The rs_etphy5000r_lnC power supplies in the l of SoC.
Refer to the check method	All connection betwee rs_etphy5000r_ln05lpe input ports of SoC, wh
Refer to the check method	In order to prevent CD diodes are inserted in connected to SoC grou vias are highly recomr
Is minimum distance between IP and the other glue logic to ensure no DRC violation secured? What is the distance between IP and noisy or high power consuming IPs (e.g. DDRPHY)?	-. IP Layout guide does however, DRC rule s Space between IP ar Place the noisy outp Please specify the dist
Routing guide at IP Side	There are signals and ring.
Check if all devices in IP are available in the product.	Please check device lis - In IP BE-Common, LV Product designer shou contact IP designer. - Product designer sha
Is the ground of the IP merged with ground plane of SOC in package level?	Please review the grou material

package's power configuration for IP	Package should be des for vdda_d (also know clean supply) may not shorted in dedicated v
Is the package sign-off confirmed by DT(Design technology) and package team?	Please review and atta
Are ballmaps for IP reviewed with IP designer? If there is an fixed signal order of connector, are signal balls correctly placed?	rs_etphy5000r_ln05lpe illustrates the bump pl minimum bump pitch pls refer to the 1.3.2 b Please attach ball-map If that IP use connecto

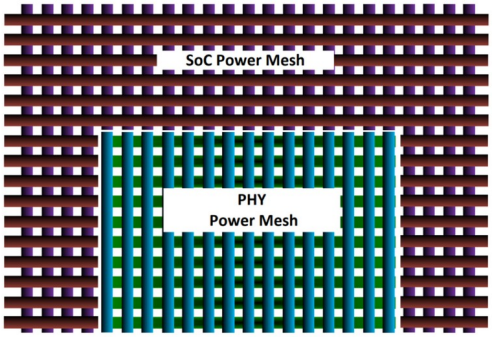
COLOR DESCRIPTION (ITEM)	
Item	PASS
For Project Manager or SOC (Block Top)	NG
For Logical Designer(RTL)	N.A
For Physical Designer(Layout - Physical Design)	Not yet
For Physical Designer(Physical Implementation)	SCORE

CHECK METHOD	CHECK RESULT
ersion, LVS/DRC signoff, and BE design guide compliance.	Not yet
elease info and internal version control logs.	Not yet
compliance and simulation/ATE verification for all SGMII modes.	Not yet
ulation and specification to confirm duplex handling.	Not yet
ntation and waveform dump for TX/RX SerDes path.	Not yet
, lock time simulation, and stability check under PVT variation.	Not yet
k diagram and simulation coverage on encoding/decoding, auto-negotiation, and link	Not yet

ısfer with MAC and APB control behavior through simulation/STA.	**Not yet**
ration test (with external PHY), SGMII waveform view, and STA analysis on I/O timing.	**Not yet**
k mux control condition, and simulation to confirm correct input selection.	**Not yet**
ncy, duty cycle, and jitter analysis result.	**Not yet**
;wing, eye diagram (post-layout), and STA/EMIR clock tree analysis.	**Not yet**
ıments onment (mobile, automotive, network) e condition ycle(time)	**Not yet**
est version of DK from Rasmschip? used DK and released date in column J, 'Note'	**Not yet**
file path and file name s hierarchical information vaveform view & version	**Not yet**
e of the power sequence or dump file. file, please share file path / file name / IP's hierarchical information. r dump file, please make comments about power sequence(e.g. ALWAYS ON, ALIVE,	**Not yet**
t the clock input and control signals are applied according to the start-up sequence ;heet? ;equence tap in this excel sheet"	**Not yet**
t the power sequence applied in the product follows the power sequence described in	**Not yet**
ıere is the case that input control signals change while the IP is enabled and in normal y problem? Have you simulated the operation scenario with the verilog model in the nment?	**Not yet**
file path and file name s hierarchical information vaveform view & version	**Not yet**
hich outputs the target frequency after a certain time, have you checked whether this :em implementation where the system waits for the lock time/settling time as specified	**Not yet**

t, in the test mode for evaluating the IP's characteristics, the maximum frequency the specified maximum operating speed of I/O? equency output in column J 'Note'	**Not yet**
attach confirmation email from IP designer	**Not yet**
file path and file name s hierarchical information on fullchip vaveform view & version	**Not yet**
here is any case where specific inputs of IP are connected to a ROM code or tie cell and fied by SW modification? t of IP's inputs connected to ROM code or signal in column J, 'Note'	**Not yet**
cases when IP is connected to ROM/Tie cell by Verilog simulation and checked there is	**Not yet**
nlable, please write "All pins are used and controllable" with the name of the person in ns should be connected to the ground (in case of the active high wer (in case of the active low operation). pins should be the floating net (NC).	**Not yet**
been set up correctly? PSEL, and PWDATA signals are modified after the rising edge of the PCLK. g edge of the PCLK, PENABLE signal is asserted. and PSEL signals remain valid throughout the access cycle and the transfer completes . deasserted at the end of the transfer. The PSEL signal proceeds to low. hould be set when reading or writing register through APB signals.	**Not yet**
been set up correctly? and PSEL signals are modified after the rising edge of the PCLK. g edge of the PCLK, PENABLE signal is asserted. s the PRDATA during the access cycle. ed on the rising edge of PCLK at the end of the access cycle.	**Not yet**
ort of the time delay from GPIO to target ports of IP (This is for checking timing skew) post STA, about IN/OUT multi-bit. uide to check target pins and timing requirements.	**Not yet**
eport: Refer to Datasheet & Userguide to check IP's pin name of interest with respect to onstraint path list	**Not yet**

clock should be tightly controlled to ensure very high-speed clock operation and its tter performance of rs_etphy5000r_ln05lpe is dependent on a low-jitter reference clock clock-tree for the clock signal distribution in full-chip level, the additional jitter from this ould be minimized (e.g. Period jitter should be very small at the worst-case simulation the jitter performance can be degraded due to low quality input reference clock. to have a review session with PHY design team for the reference clock scheme and the lock jitter.	**Not yet**
lt of the input clock jitter simulation from the clock source on SOC to the IP's input clock k jitter with clock jitter simulation flow including on-chip network tape-out. it must be checked through integration review flow.	**Not yet**
e should be placed in the center of one-side to minimize package R/L/C parasitic. If it e center, there should be enough space (e.g. > 1000um) from the chip edge. mmended to be far away from noisy switching logic. The distance to other IP and logic d as 20µm and 40µm, respectively ks such as DDR PHY, then place this IP at the other side of the die (e.g. the opposite ing IO signals from the DDR PHY may cause unwanted noise coupling in the die and/or To prevent unwanted notice, ensure that you place the noisy block at the opposite side oring block placement, at least one block should be located near the PHY. We ce the process monitoring block within 100um from the PHY and closer to the IO region um from chip edge). This block can be located with the minimum space rule of 5um ~ 300 um space → 5 um space) because this block is not working in the normal mode.	**Not yet**
t appropriate type of I/O are used? Write the name of the used I/O in column J, 'Note'	**Not yet**
e IP properly separated from noise interference around using slot cells? Have you D issues with ESD team?	**Not yet**
t there is no noisy sources (i.e. Tx Driver/Signal I/O) placed around the IP's Power I/O?	**Not yet**

wer supply noise which induces jitter on the TX and RX waveforms, e's power and ground should be isolated from those of the SoC as illustrated below  	**Not yet**
)5lpe Hard Macro includes primary ESD protection devices for all signal pads and all PHY. Internal ESD protection devices are not required to be connected to any power pin	**Not yet**
n the PHY and SoC requires ESD protection using CDM diodes. All input ports of e have internal CDM diodes but all output ports do not have the diodes. Therefore, all ich are connected to rs_etphy5000r_ln05lpe's output, must have CDM diode.	**Not yet**
M damage between rs_etphy5000r_ln05lpe and SoC power supplies, back-to-back the PHY. The "vsscore(Vbb)" ports on rs_etphy5000r_ln05lpe should be strongly unds as illustrated in Figure 4. Thicker metal layers like Kx or Gx with enough number of nended for the "vsscore(Vbb)" routing.	**Not yet**
s not limit minimum space between IP and the ohter glue logic. hould not be violated. 5-um space is recommended for DRC rule. nd the noisy IO output buffer(for example DDR PHY) is important. ut pad as far as possible from IP. ance or attach a captured image	**Not yet**
powers at IP side. signal or power routing with these ports is not allowed except the I/O	**Not yet**
st with this method in below. 'S report(lvsprt_ca.lvs) file includes IP device list. ld check if all of the devices are available in the product. If there is an issue, please re the device list of product, then IP designer would check it.	**Not yet**
und ball name in ball map and where the merging point is and attach the confirm	**Not yet**

signed to dedicate a plane for vssa. It is also recommended to have a dedicated plane n as dirty supply) to minimize the package inductance; while vdda_c (also known as have dedicated plane but must have spate ball out. The vddh_ln and vddh_cmn can be ddh (1.8V) plane but it is recommended to have multiple ball outs for this domain.	**Not yet**
ach the results	**Not yet**
e is designed to support only flip chip package, but not wire-bond type. Figure 2 lacement. No bump is required to be added in SoC. The bump size is 71um, and the is 150um in the below bump map. ump map (rs_etphy500r_ln05lpe_userguide) os and review confirmation emails from IP designer. r, please fill in signal order of connector	**Not yet**

ram

SCORE		
0	PASS	
0	NG	
0	N.A	
48	Not yet	
0%	SCORE	3(

PRODUCT (PROJECT)		IP	
NAME/DATE (YY.MM.DD)	NOTE	CHECK RESULT	NAME/DATE (YY.MM.DD)
		Not yet	
		Not yet	
		PASS	rice & issac / 25.09.22
		Not yet	
		PASS	rice & issac / 25.09.22
		Not yet	
		PASS	rice & issac / 25.09.22

		PASS	rice & issac / 25.09.22
		Not yet	
		Not yet	
		Not yet	
		Not yet	
		Not yet	
		Not yet	
		PASS	rice & issac / 25.09.22
		PASS	rice & issac / 25.09.22
		PASS	rice & issac / 25.09.22
		Not yet	
		PASS	rice & issac / 25.09.22
		PASS	rice & issac / 25.09.22
		PASS	rice & issac / 25.09.22

		Not yet	
		PASS	rice & issac / 25.09.22
		Not yet	
		N.A	
		N.A	
		PASS	rice & issac / 25.09.22
		PASS	rice & issac / 25.09.22
		PASS	rice & issac / 25.09.22
		Not yet	
		Not yet	

		Not yet	
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		Not yet	
		Not yet	
		Not yet	
		Not yet	

		Not yet	
		Not yet	
		Not yet	

-sCHIP
GIXEDIGNAL

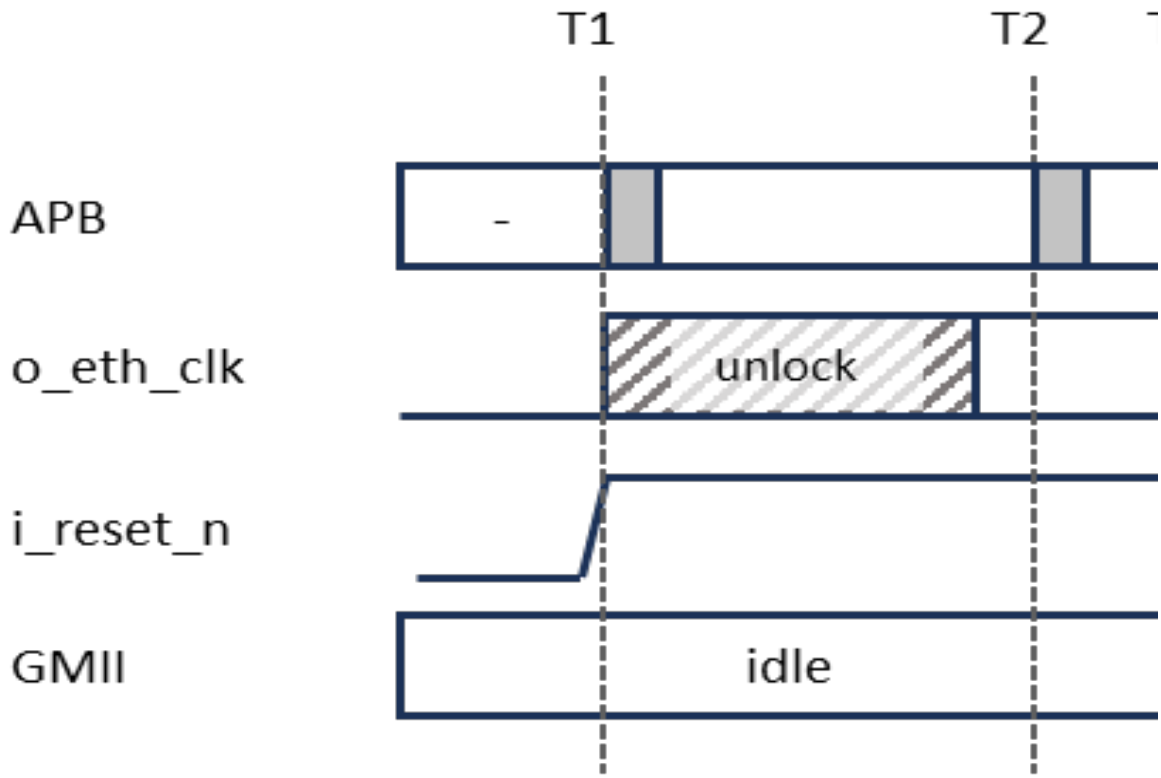
14
0
2
32
0%

VER. 1.0

NOTE
to do half duplex

Not yet external 156.25 MHz input
Use Internal Loopback Test in aging phase

POWER UP & Initialization

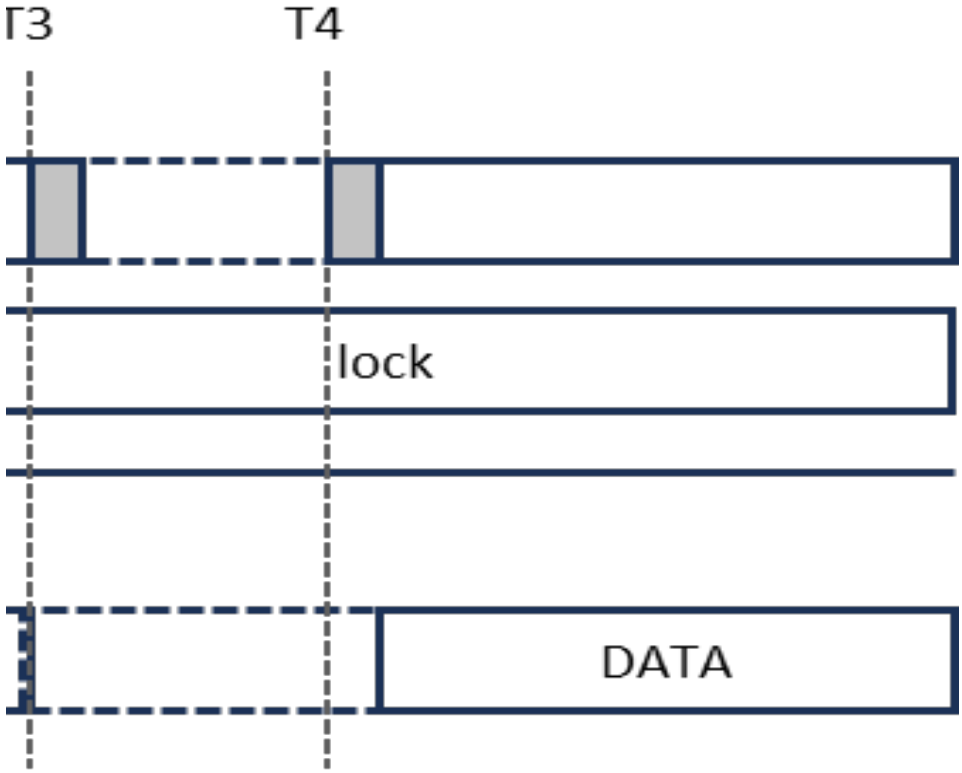


Power up the VDD/VDIG/VDDT before VDDH.

T1: Assert reset(i_reset_n) And configure the initial setting of P(
T2: After the clock lock is completed, configure PCS and PMA to
T3: If Auto Negotiation is enabled at T1, Check the the “an_com
interrupt.
T4: If Auto Negotiation is enabled at T1, Reconfigure the speed

POWER DOWN SEQUENCE

The power supplies of VDD/VDIG/VDDT/VDDH may be powered



CS and PMA
start.
nplete” flag by polling the apb or using an

according to the AN results

down in any order.